Report: Differential Gene Expression Analysis of Blood Samples from Patients with Chronic Spontaneous Urticaria

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Background

Chronic spontaneous urticaria (CSU) is a skin disease characterized by intense itching, wheals or hives, and angioedema (deep tissue swelling) that recur for more than six weeks. Unlike allergic urticaria, CSU is not triggered by external allergens and is often associated with autoimmune mechanisms in which the immune system stimulates mast cells to release histamine. In this report, differentially expressed genes (DEGs) in the blood of patients with CSU were identified to better understand the molecular mechanisms underlying the pathophysiology of CSU.

Methods

The dataset GSE72541 was obtained from the Gene Expression Omnibus (GEO) database of the National Center for Biotechnology Information (NCBI). The samples consisted of blood from patients with CSU and healthy individuals as controls. Gene expression profiling was performed using the Agilent SurePrint G3 Human Gene Expression v2 8×60K Microarray (GPL16699). The dataset originates from a study conducted by Giménez-Arnau et al. (2017).

Data analysis was performed using RStudio. Required packages were installed using BiocManager, including GEOquery, limma, and AnnotationDbi from Bioconductor, as well as pheatmap, ggplot2, dplyr, and umap from CRAN.

The dataset was retrieved using GEOquery, and gene expression values were log2-transformed for normalization. Samples were grouped into CSU patients and controls, and a design matrix was constructed for differential gene expression analysis using the limma package.

Differentially expressed genes were then annotated and visualized using boxplots, density plots, UMAP, volcano plots, and heatmaps to evaluate data distribution and expression patterns. Finally, significant genes were subjected to Gene Ontology (GO) and KEGG pathway enrichment analysis using g:Profiler to identify biological processes and pathways associated with CSU.

Result and Interpretation

The volcano plot illustrates the distribution of genes based on their statistical significance and magnitude of expression change. In this analysis, the threshold parameters used were adjusted p-value (adj. p. val) < 0.05 and $|\log FC| > 0.1$. Genes represented by red dots indicate

those that are significantly upregulated in the CSU group. In contrast, blue dots represent genes that are significantly downregulated.

To identify the top upregulated and downregulated genes, the top 20 DEGs were sorted based on the highest and lowest log fold-change (logFC) values. The results are presented in Table 1.

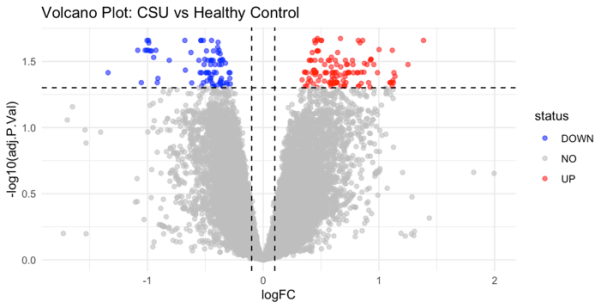


Figure 1 Volcano plot of differentially expressed genes in CSU compared to control

Table 1 Top 20 upregulated and downregulated genes in CSU compared to control

Upregulated genes			Downregulated genes		
logFC	**adj.P.Val**	**SYMBOL**	**logFC**	**adj.P.Val**	**SYMBOL**
1,3858	0,0219	ITGB3	-1,3408	0,0385	NEURL
1,2516	0,0334	AQP10	-1,0839	0,0260	HLA-DRB3
1,1392	0,0410	VWF	-1,0530	0,0457	PID1
1,1158	0,0260	ITGA2B	-1,0204	0,0260	CXCR6
1,1150	0,0488	VWF	-0,9155	0,0457	FAM26F
1,1093	0,0457	MXD1	-0,9099	0,0425	ALDH1A1
1,0015	0,0257	PRKAR2B	-0,6764	0,0219	NETO2
0,9879	0,0385	QKI	-0,6726	0,0368	NID1
0,9560	0,0334	RAPGEF2	-0,6243	0,0271	IRF8
0,9334	0,0305	SAMD14	-0,6174	0,0460	P2RY2
0,9302	0,0305	THBS1	-0,5559	0,0310	LOC375196
0,9257	0,0310	TAB2	-0,5451	0,0385	SLC25A5
0,9238	0,0494	CMTM5	-0,5386	0,0219	SAP30
0,9108	0,0457	GRB10	-0,5340	0,0487	CYCS
0,8974	0,0334	C6orf25	-0,5329	0,0385	FOLR2
0,8749	0,0327	AFF1	-0,5171	0,0457	TMEM18
0,8708	0,0381	CLU	-0,4957	0,0385	FAM86B2
0,8597	0,0219	GP1BB	-0,4884	0,0385	MFSD2A
0,8486	0,0381	GP9	-0,4863	0,0445	CNIH
0,8475	0,0463	ENTPD1	-0,4608	0,0420	MTHFD2

The heatmap displays the top 50 differentially expressed genes between the CSU and control groups. The visualization shows a clear and consistent clustering pattern, where CSU samples and control samples form distinct clusters.

In the heatmap, the color gradient represents gene expression levels, with red indicating higher expression and blue indicating lower expression. The contrasting color patterns between the two groups highlight substantial differences in gene expression profiles, suggesting that the genetic expression patterns of CSU patients differ markedly from those of healthy individuals.

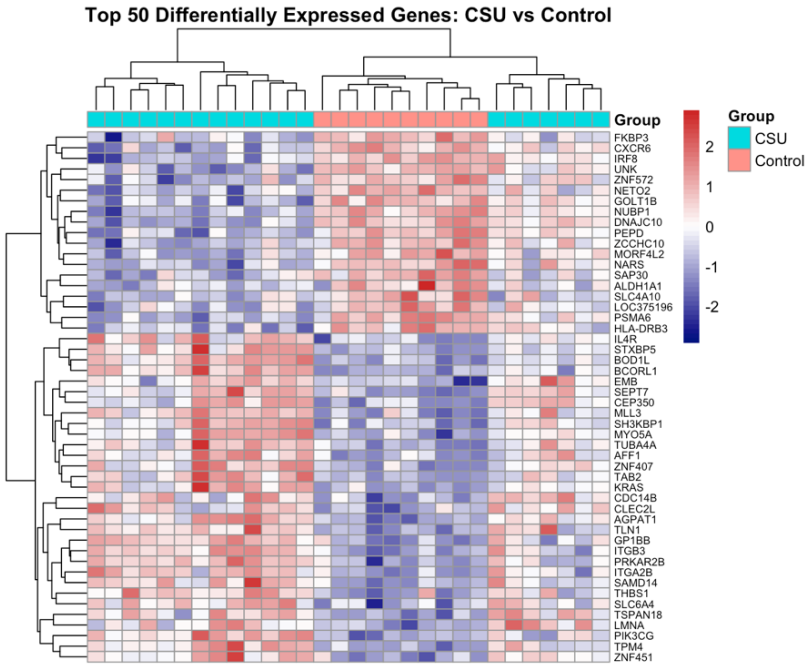
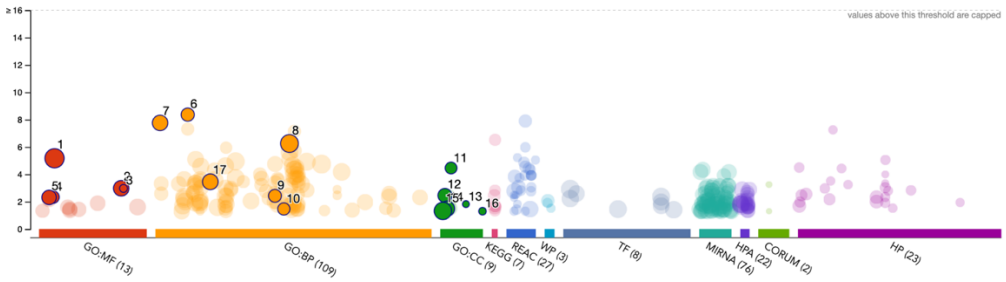


Figure 2 Heat map of the top 50 differentially expressed genes (DEGs) in CSU compared to control

Further analysis was conducted through Gene Ontology (GO) enrichment analysis using g:Profiler to better understand the biological roles of the identified genes in the pathogenesis of CSU.

Based on the results summarized in Figure 3, two major biological process groups were highlighted. The first involves hemostasis and platelet function, where significant involvement was observed in blood coagulation and cellular activation processes. This finding suggests that platelets may play an important role in the inflammatory response associated with CSU. The second group involves the regulation of the adaptive immune system, particularly pathways related to T cell activation and differentiation. These findings support the hypothesis that CSU may be associated with autoimmune mechanisms.

Overall, these results suggest that CSU may be characterized by a pro-coagulant state and systemic immune activation. The strong signals associated with platelet activity and hemostasis indicate that platelets may function as inflammatory cells, releasing mediators that link coagulation processes with the development of urticaria symptoms that resemble allergic reactions.



Source	Term ID	Term Name	p_{adj} (query_1)
GO:MF	GO:0005515	protein binding	6.712×10^{-6}
GO:MF	GO:0060090	molecular adaptor activity	1.053×10^{-3}
GO:MF	GO:0070051	fibrinogen binding	1.070×10^{-3}
GO:MF	GO:0005178	integrin binding	4.616×10^{-3}
GO:MF	GO:0004674	protein serine/threonine kinase activity	4.871×10^{-3}
GO:BP	GO:0007599	hemostasis	4.334×10^{-9}
GO:BP	GO:0001775	cell activation	1.736×10^{-8}
GO:BP	GO:0048523	negative regulation of cellular process	5.582×10^{-7}
GO:BP	GO:0045580	regulation of T cell differentiation	3.755×10^{-3}
GO:BP	GO:0046634	regulation of alpha-beta T cell activation	3.416×10^{-2}
GO:CC	GO:0031091	platelet alpha granule	3.465×10^{-5}
GO:CC	GO:0005925	focal adhesion	3.511×10^{-3}
GO:CC	GO:0070442	integrin alphaIIb-beta3 complex	1.508×10^{-2}
GO:CC	GO:0009986	cell surface	3.263×10^{-2}
GO:CC	GO:0005654	nucleoplasm	4.749×10^{-2}
GO:CC	GO:1990779	glycoprotein Ib-IX-V complex	5.000×10^{-2}
GO:BP	GO:0016310	phosphorylation	3.522×10^{-4}

rsion	e113_eg59_p19_6be52918
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ganism	hsapiens

g:Profiler

Figure 3 Enrichment analysis of Gene Ontology (GO) terms for upregulated genes in CSU

Further functional interpretation was performed through KEGG pathway enrichment analysis to identify molecular pathways associated with the differentially expressed genes. The analysis revealed significant enrichment in pathways related to platelet function, particularly the platelet activation pathway. This pathway involves several keys signaling molecules, including PI3K, PKC, and Rap1, which play important roles in intracellular signal transduction within platelets.

Activation of this pathway triggers two major processes: platelet aggregation and degranulation. The degranulation process leads to the release of various inflammatory mediators, which may mechanistically contribute to the development of clinical symptoms such as urticaria (hives) and angioedema in patients with CSU. These findings suggest that platelet activation plays an important role in the pathophysiology of CSU.

In addition, the KEGG analysis also identified significant involvement of the PI3K–AKT signaling pathway, which showed increased systemic activity. The upregulation of this

pathway indicates two major biological implications. First, it suggests platelet hyper-responsiveness, where elevated PI3K–AKT activity may enhance the pro-coagulant state observed in patients. Second, it is associated with sustained inflammatory responses, as this pathway is also involved in T-cell circulation and activation, contributing to the chronic inflammatory profile characteristic of CSU.

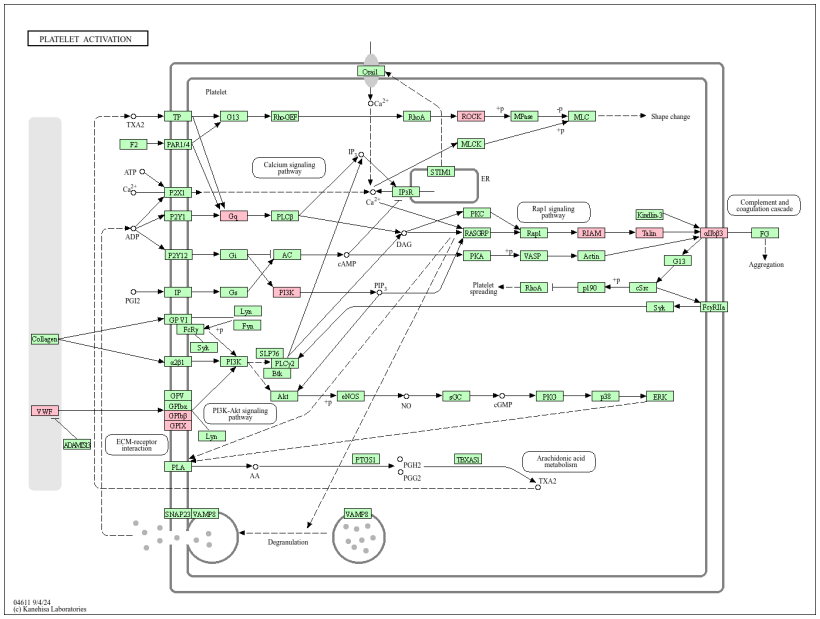


Figure 4 KEGG Platelet Activation pathway map with key genes highlighted in red

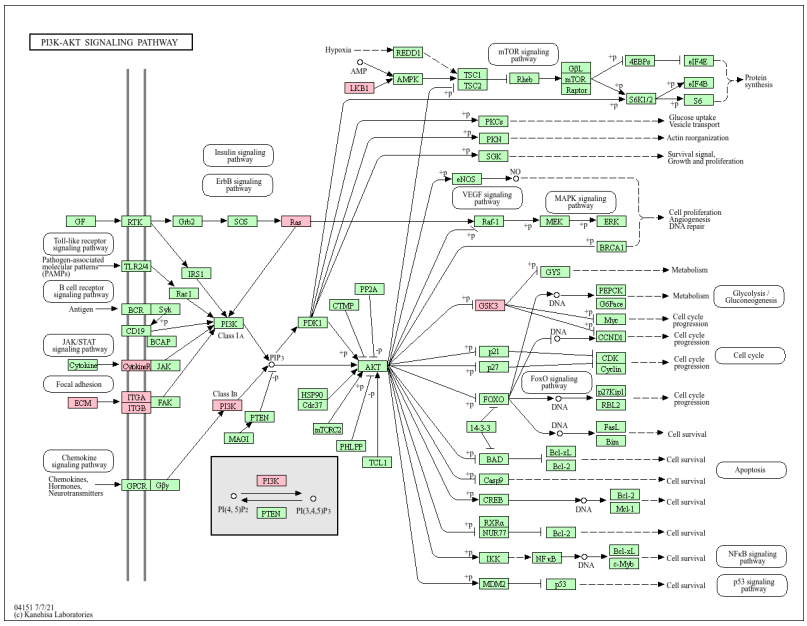


Figure 5 KEGG PI3K-Signaling pathway map with key genes highlighted in red

Conclusion

Based on the differential gene expression analysis of patient blood samples, the pathophysiology of CSU appears to be associated with the upregulation of genes involved in platelet function and blood coagulation processes.

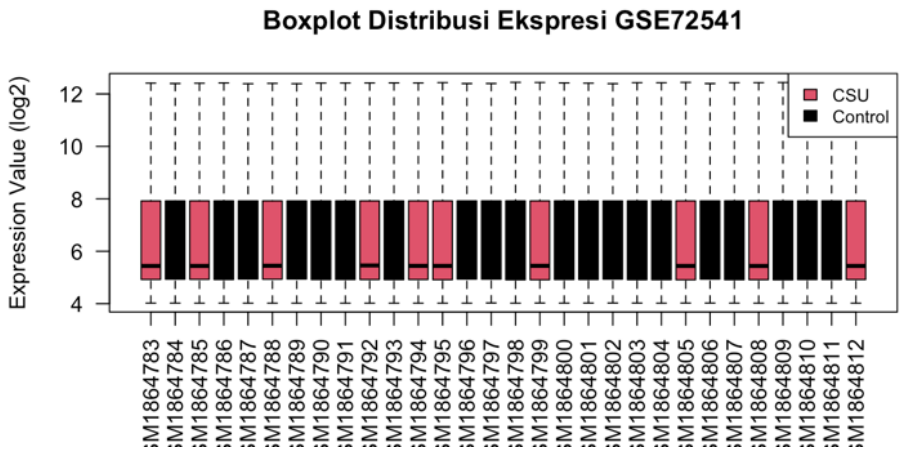
In addition, the analysis also revealed the involvement of adaptive immune system activation and increased activity of the PI3K–AKT signaling pathway, which may contribute to the pro-coagulant state and systemic inflammation observed in patients.

Overall, these findings suggest that the interaction between the coagulation system and the immune system represents an important mechanism underlying the pathophysiology of CSU.

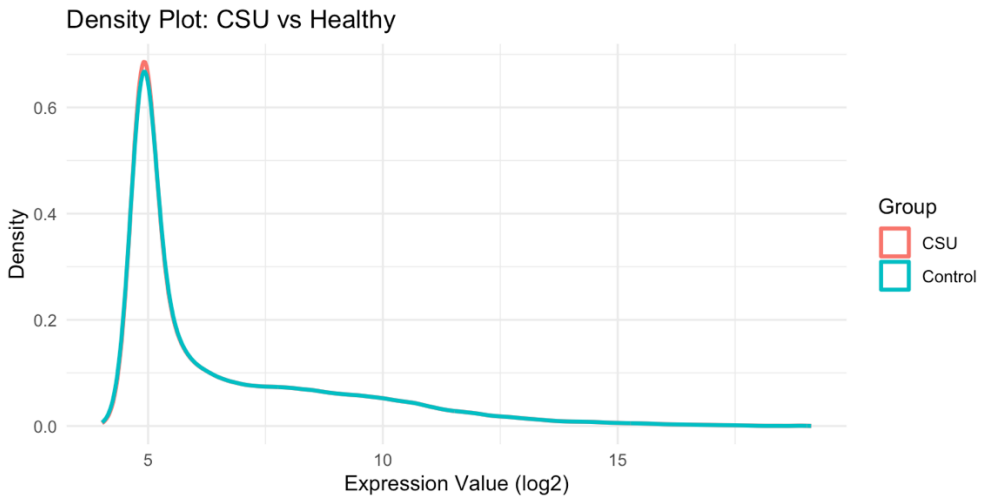
References

Giménez-Arnau, A., Curto-Barredo, L., Nonell, L., Puigdecanet, E., Yelamos, J., Gimeno, R., Rüberg, S., Santamaria-Babi, L., & Pujol, R. M. (2017). Transcriptome analysis of severely active chronic spontaneous urticaria shows an overall immunological skin involvement. Allergy, 72(11), 1778–1790. https://doi.org/10.1111/all.13183

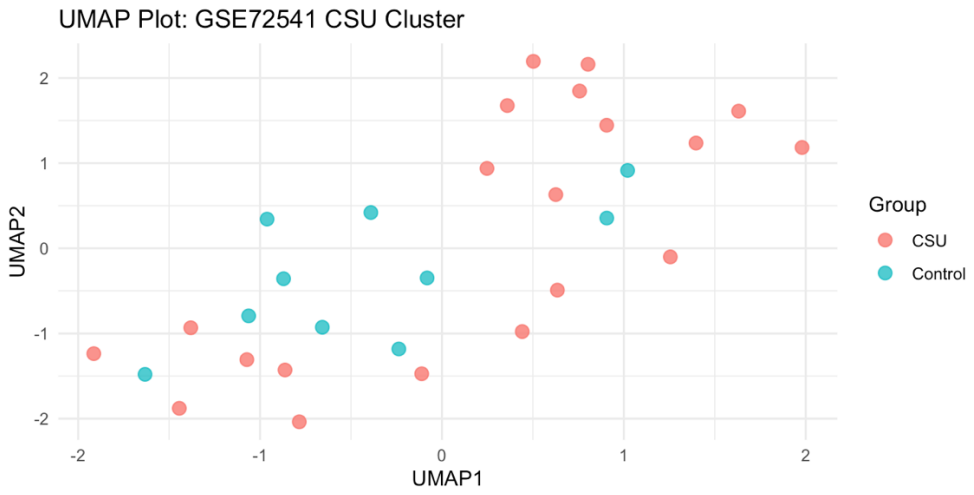
Appendix



Appendix 1 Boxplot of expression value distribution per sample



Appendix 2 Density plot of gene expression value distribution



Appendix 3 UMAP plot of CSU and control samples